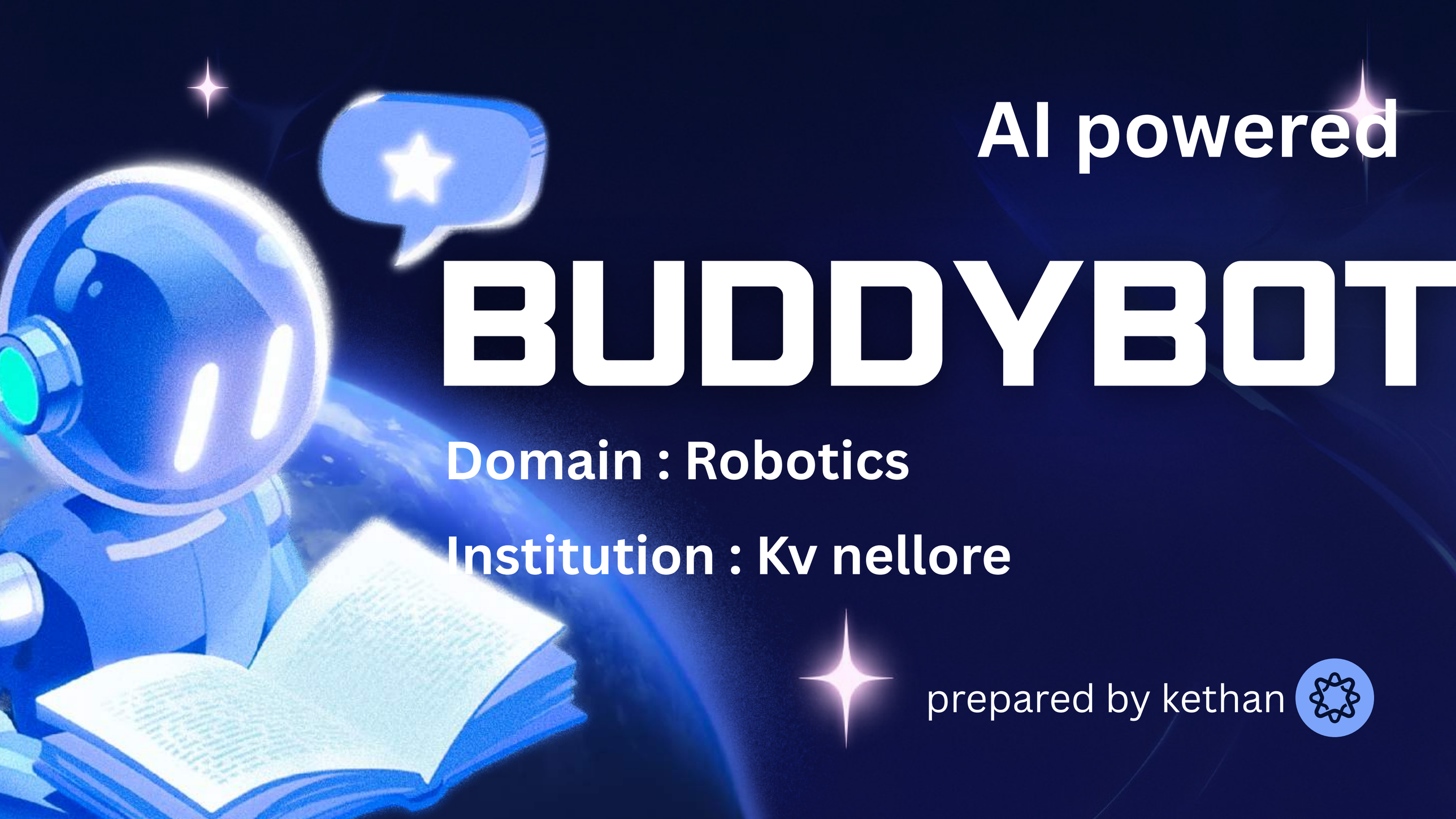

AI powered
BUDDYBOT
Domain : Robotics
Institution : Kv nellore
prepared by kethan

ABSTRACT

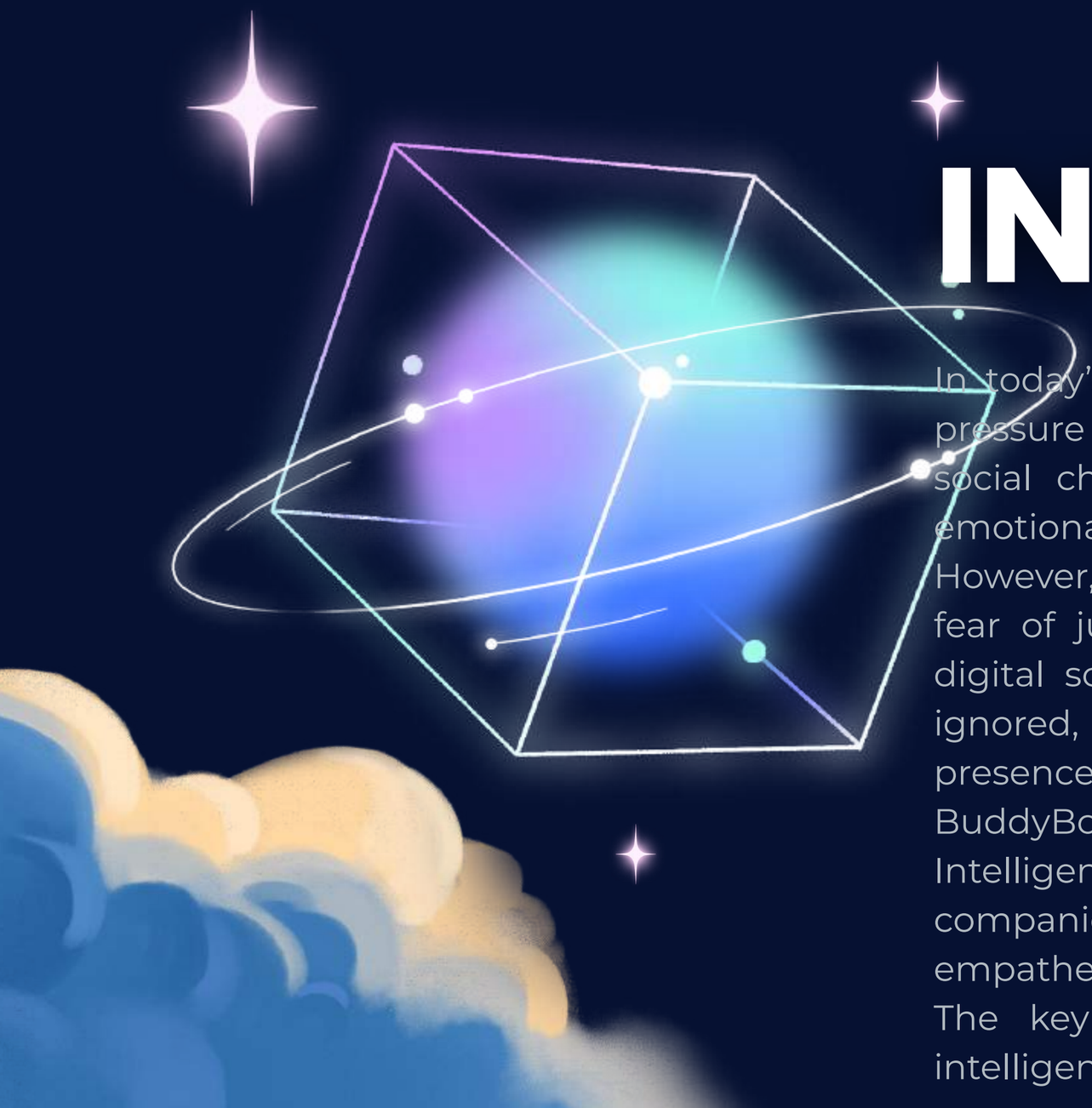
BuddyBot is an advanced AI-powered emotional companion robot designed to assist students in managing stress, anxiety, loneliness, and emotional instability. It integrates Artificial Intelligence, Natural Language Processing, Speech Recognition, Text-to-Speech systems, and embedded robotics into a unified interactive system.

Unlike traditional software-based chatbots, BuddyBot is a physical desktop robot that creates a sense of presence and emotional connection with the user. It continuously listens for wake words, processes voice input, analyzes emotional tone, and responds in a human-like conversational manner.

The system also includes a backend architecture that stores emotional trends, session logs, and behavioral patterns, allowing long-term emotional analysis. Additionally, it incorporates a multi-model AI switching mechanism to ensure uninterrupted service during API failures or network issues.

BuddyBot is designed as a low-cost, scalable, and intelligent emotional support system for students, aligning with modern AI-driven mental health solutions.





INTRODUCTION

In today’s academic environment, students face increasing levels of pressure due to competition, examinations, parental expectations, and social challenges. These factors often lead to stress, anxiety, and emotional imbalance.

However, most students do not express their emotions openly due to fear of judgment or lack of a safe communication space. Existing digital solutions such as mobile applications or chatbots are often ignored, lack emotional depth, and do not create a sense of human presence.

BuddyBot was developed to bridge this gap by combining Artificial Intelligence with a physical robotic system. The robot acts as a “digital companion” that listens, understands emotions, and responds empathetically.

The key innovation is not just AI conversation, but emotional intelligence + physical interaction + continuous monitoring system.

PROBLEM STATEMENT

ACEDEMIC PRESSURE

- High competition in exams
- Fear of failure
- Parental expectations

EMOTIONAL CHALLENGES:

- Stress and burnout
- Loneliness and isolation
- Lack of emotional communication

LIMITATIONS OF EXISTING SOLUTIONS:

- Mobile apps are easily ignored
- Chatbots lack emotional depth
- No physical interaction system
- No continuous emotional tracking

There is no affordable, physical, AI-based companion system that can continuously interact with students and detect emotional states in real time.
BuddyBot solves this problem by combining:

- AI conversation system
- Emotion detection pipeline
- Physical robotic presence
- Real-time feedback system

OBJECTIVES

The primary objectives of BuddyBot are:

1. **Emotional Detection**

To analyze student emotions using AI-based sentiment analysis and classify emotional states.

2. **Human-Like Interaction**

To create a natural conversational experience using speech recognition and text-to-speech systems.

3. **Continuous Monitoring**

To track emotional changes over time and store data securely.

4. **Early Warning System**

To detect high-risk emotional states and trigger alerts.

5. **Parent/Teacher Insights**

To provide analytical dashboards showing emotional trends.

6. **Low-Cost Implementation**

To ensure affordability for educational institutions



SYSTEM ARCHITECHTURE



BuddyBot is built using a layered architecture system:

- Hardware Layer
  - Microphone (voice input capture)
  - Speaker (audio output)
  - Desktop robotic structure
  - USB power system (5V DC)
- Software Layer
  - Python-based AI engine
  - Flask backend server
  - Emotion detection pipeline
  - Speech recognition module
  - Text-to-speech system
- Data Layer
  - SQLite database
  - Session storage system
  - Emotional trend logs
  - Analytics data structure
- Intelligence Layer
  - Multi-model AI switching system
  - Fallback response engine
  - Mood severity classification system
  - Behavioral analysis module

REAL WORLD APPLICATIONS

BuddyBot works in a continuous loop system:
Step 1: Wake Word Detection
Microphone listens for activation phrases like:

- “Hey Buddy”
- “Hello”
- “Hi”

Step 2: Voice Capture
User speech is captured in real time.
Step 3: Speech-to-Text Conversion
Audio is converted into text using speech recognition models.
Step 4: Emotion Analysis
Python-based NLP pipeline analyzes:

- sentiment polarity
- emotional keywords
- stress indicators

Step 5: AI Response Generation
AI model generates emotionally appropriate responses.
Step 6: Text-to-Speech Output
Response is converted into natural speech.
Step 7: Data Logging
All interactions are stored in SQLite database.





Backend Framework:
BuddyBot uses Flask as a lightweight backend framework.
Why Flask:

- Minimal overhead
- Fast API routing
- Suitable for real-time processing
- Easy integration with Python AI modules

APIs Used:

- /listen → captures user input
- /analyze → processes emotion
- /dashboard → displays analytics

Multi-Threading System:

- Separate thread for audio input
- Separate thread for AI processing
- Prevents system freezing

AI Switching Logic:
If one model fails:

- system switches to backup AI model
- ensures uninterrupted conversation

SOFTWARE ENGINEERING DETAILS



FUTURE OUTLOOK

BuddyBot uses a custom emotional intelligence system:

Emotion Categories:

- Normal
- Medium stress
- High-risk emotional state

Detection Process:

- Text sentiment analysis
- Keyword scanning
- Context understanding
- Rolling average emotional score

Severity Classification:

- NORMAL → stable condition
- MEDIUM → mild stress
- HIGH → emotional risk alert

Advanced Feature:

- Multi-session emotional tracking
- Behavioral trend detection

HARDWARE DESIGN

BuddyBot hardware consists of:

Components:

- Microphone array for voice capture
- Speaker for AI output
- Lightweight acrylic/PLA body
- USB power supply

Design Goals:

- Compact desktop structure
- Easy assembly
- Low power consumption
- Safe for school environments

Engineering Constraints Solved:

- Echo cancellation using software mute lock
- Noise filtering for classroom environments
- CPU optimization for continuous listening



INNOVATION & FEATURES

BuddyBot introduces multiple innovations:

Core Features:

- Real-time voice interaction
- Emotion detection system
- AI conversational intelligence
- Wake word activation
- Behavioral analytics

Advanced Features:

- Multi-model AI switching
- Offline fallback system
- Parent monitoring dashboard
- Emotion trend visualization

Key Innovation:

Combination of:

AI + Robotics + Emotional Intelligence + Monitoring System



APPLICATIONS

BuddyBot can be used in:

- Schools
- Colleges
- Hostels
- Counseling rooms
- Home environments

Use Cases:

- Stress management
- Emotional monitoring
- Student counseling support
- AI learning assistant
- Mental wellbeing tracking

THANK YOU!

